UVA | SCHOOL of DATA SCIENCE

# Extensions to Q-Learning 2: Dueling Deep Q-Network

Reinforcement Learning

School of Data Science

University of Virginia

Last updated: October 17, 2026

# Dueling Networks

Deep RL before this paper used conventional architectures (CNN, LSTM)

Focus here is new architecture better suited to model-free RL

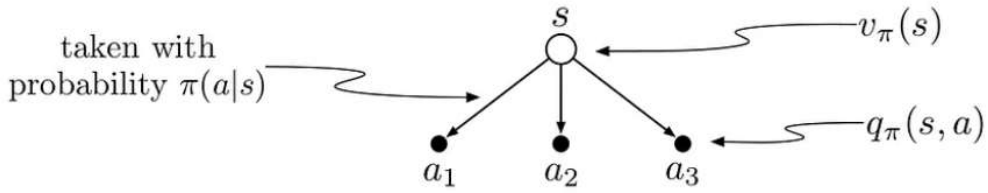
*Dueling architecture* separates state values and action advantages

Paper: *Dueling Network Architectures for Deep Reinforcement Learning.* Wang et. al.

# Value and Advantage Function Definitions

Relationship of value functions:

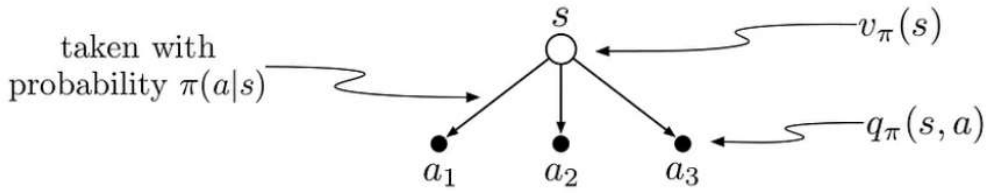
$$
\begin{aligned}
Q^{\pi}(s,a) &= \mathbb{E}\left[R_t \middle| s_t = s, a_t = a, \pi\right] \\
V^{\pi}(s) &= \mathbb{E}_{a\sim\pi(s)}\left[Q^{\pi}(s,a)\right].
\end{aligned}
$$



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The advantage function isolates effect of action taken

$$A^{\pi}(s,a) = Q^{\pi}(s,a) - V^{\pi}(s)$$

V(*s*) measures how good it is to be in state *s*

A(s,a) measures relative importance of each action

Sometimes, the action doesn't matter

# Dueling Architecture

Separate streams for value and advantage functions

Common convolution feature

Top figure is Q-network

Bottom figure is dueling Q-network

Value and advantage streams are combined

For each network, outputs are Q(s,a)

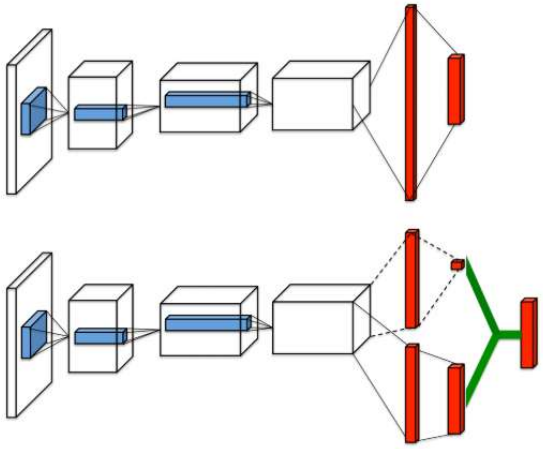
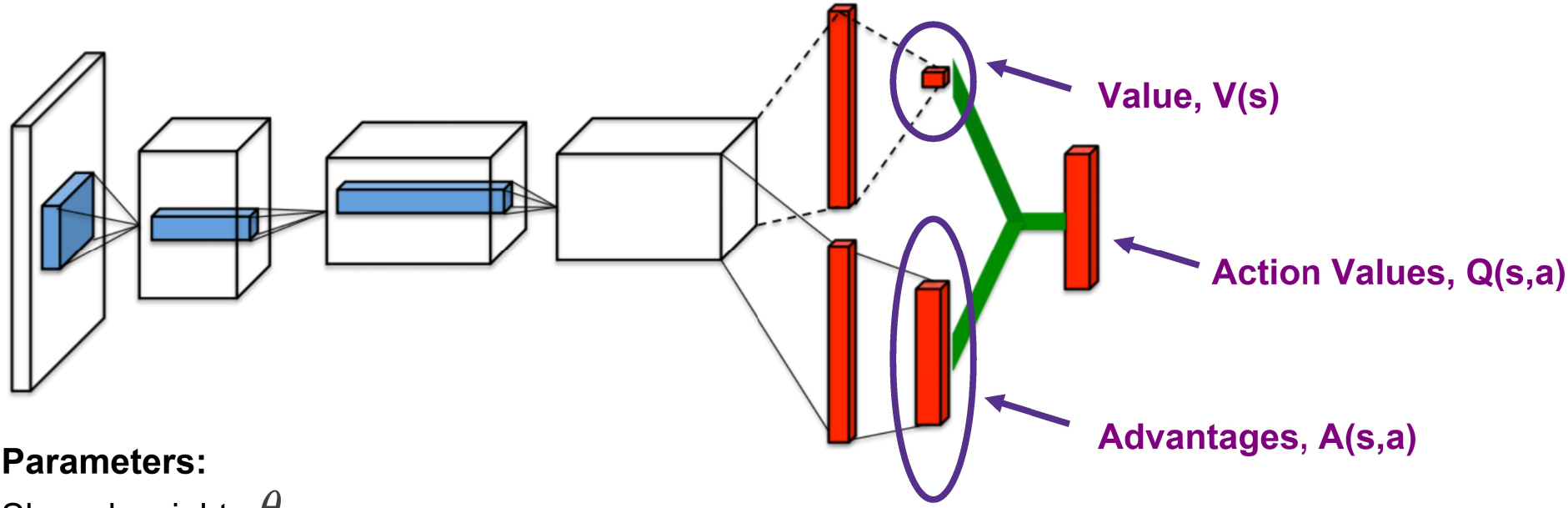


*Figure 1.* A popular single stream $Q$-network (**top**) and the dueling $Q$-network (**bottom**). The dueling network has two streams to separately estimate (scalar) state-value and the advantages for each action; the green output module implements equation (9) to combine them. Both networks output $Q$-values for each action.

# Dueling Architecture, contd.




**Parameters:**

Shared weights $\theta$

Value stream weights $\beta$

Advantage stream weights $\alpha$

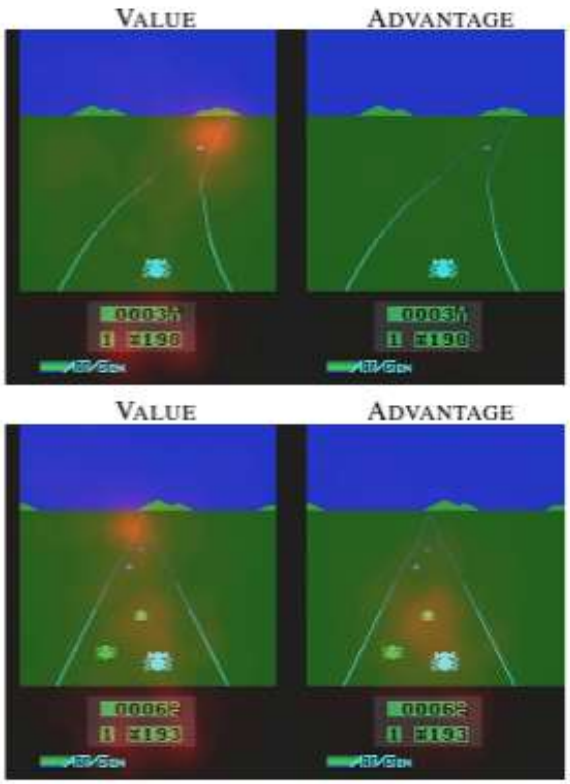
# Value and Advantage Function Saliency

Figure shows saliency map for two time steps

Value stream pays attention to horizon and score

Advantage stream: when no cars are on road, action doesn't matter (no attention paid)

When cars are on road, advantage stream pays attention to car in front (bottom right)



# Identifiability

Given Q = V + A, we cannot recover V and A uniquely
(adding any *c* to V, and subtracting *c* from A, leaves Q unchanged)

Add a constraint: subtract the average advantage from A:

$$Q(s,a;\theta,\alpha,\beta) = V(s;\theta,\beta) + \left( A(s,a;\theta,\alpha) - \frac{1}{|\mathcal{A}|}\sum_{a'} A(s,a';\theta,\alpha) \right)$$

For actions with above average advantage, the second term in (*) will be positive
For action with greatest advantage, (*) term will be largest across all actions

# Review of Dueling Architecture Code

We can see an implementation of Dueling Q-Network here:

Paper: *Reinforcement Learning for optimal sepsis treatment policies* (2017)

Authors: Raghu, Komorowski, Ahmed, Celi, Szolovits, Ghassemi

GitHub repo:
https://github.com/aniruddhraghu/sepsisrl/blob/master/continuous/q_network.ipynb

See section:

# advantage and value streams

# Implementation

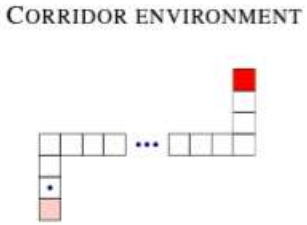
Our two streams (V, A) for Q are part of the model architecture

Training step runs the same as a standard Q-network: backpropagation

# Evaluation

Sets up simple environment called *corridor*.

Redness of state signifies reward



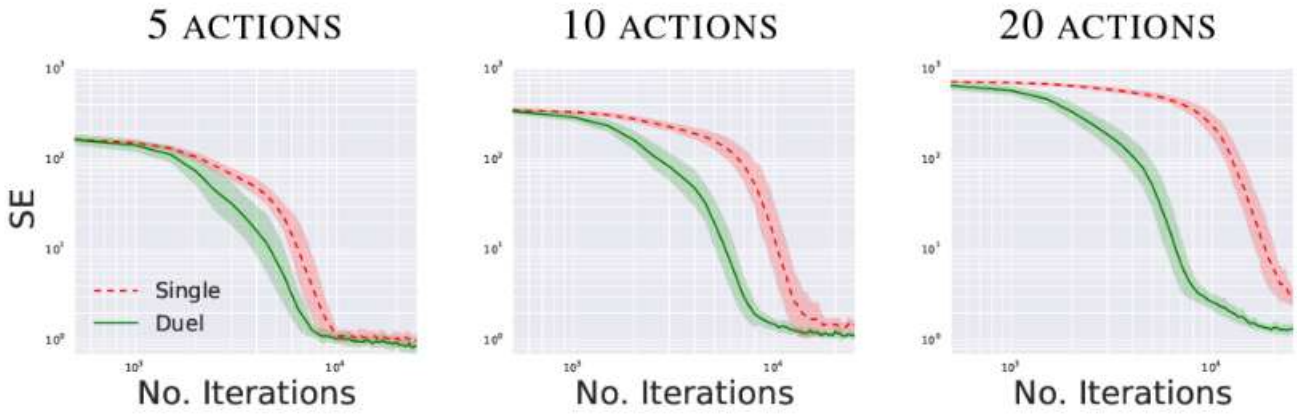

Can move up, down, left, right, or no move.

ε-greedy policy, measure performance by squared error (SE) of Q-value estimates vs. true action values

Single and Duel networks use MLP with three layers

For duel network, after first hidden layer, network branches off to two streams

# Evaluation, contd.




Dueling network converges faster than Single

More pronounced as |A| increases

As |A| increases, SE is lower for Dueling network. It performs better.

# Summary

We studied a novel architecture for Q function

Decomposes Q into value V and action advantage A streams

$$Q^{\pi}(s,a) = V^{\pi}(s) + A^{\pi}(s,a)$$

(we subtract average advantage for identifiability)

# Summary

We studied a novel architecture for Q function

Decomposes Q into value V and action advantage A streams

$$Q^{\pi}(s,a) = V^{\pi}(s) + A^{\pi}(s,a)$$

(we subtract average advantage for identifiability)

Advantage function measures incremental importance over V(s):

$$A^{\pi}(s,a) = Q^{\pi}(s,a) - V^{\pi}(s)$$

Dueling network converges faster than usual DQN, Double Q-Network

Errors shown to be lower for Atari games